A FEM-Induced Dynamic Narrow-Band Signed Distance Field with Closest-Feature Sensitivities for Finite-Element Contact

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Abstract

Signed distance fields provide efficient geometric queries for contact, but static SDFs are not directly suitable for deformable finite-element bodies whose surfaces evolve with nodal degrees of freedom. This paper presents a training-free dynamic narrow-band SDF field for FEM contact. At each time step, the current FEM boundary is used to populate a narrow-band signed distance field in the deformed configuration. In addition to distance values, each grid node stores closest-feature payloads, including face identity, barycentric coordinates, and current surface normals. These payloads allow the interpolated SDF field to provide contact gaps, scalar-field normals, and FEM-consistent master-side contact sensitivities. Contact queries are performed by field interpolation; closest-point projection is used only during field construction or optional refinement. We derive the field-based contact Jacobian using the analytic derivative of the scalar trilinear SDF interpolation, assemble frictionless penalty contact forces, and analyze the amortized cost relation between SDF field update and repeated contact queries. Validation reports field accuracy, Eikonal residuals, slave/master Jacobian finite-difference errors, and the crossover query count beyond which the dynamic SDF field outperforms projection-based contact queries.

Keywords: finite element contact; signed distance field; narrow-band method; contact mechanics; closest-feature sensitivity; surface quadrature; computational mechanics

1 Introduction

Contact in deformable finite-element simulation requires repeated evaluation of gaps, normals, contact sensitivities, and force contributions on evolving surfaces. Static signed distance fields are attractive for contact because they replace repeated geometric search with local field queries, but a static field does not remain a current-configuration distance field when a finite-element body deforms.

Existing dynamic SDF approaches often use deformation snapshots, reduced bases, learned fields, or other data-driven approximations. Those methods can be effective, but they introduce offline data requirements and may decouple the distance representation from the actual finite-element state. This work instead constructs the SDF directly from the current FEM boundary at each update.

The proposed method builds a current-space narrow-band SDF field from the current boundary surface. A spatial-hash accelerated closest-point projection kernel is used only to populate grid nodes. Once the field exists, contact queries interpolate the field and its closest-feature payload. The method therefore keeps the current-surface geometric consistency of projection methods while making the paper-facing contact query a true field interpolation.

The contributions are: a training-free FEM-induced dynamic narrow-band SDF field; closest-feature grid payloads for FEM-consistent master-side sensitivities; a variationally consistent scalar-field derivative for the slave Jacobian; field-contact integration backends covering both preserved node-to-surface sampling and surface-to-surface quadrature; and an amortized cost model with measured crossover counts.

2 Related Work

Mesh-based finite-element contact commonly defines normal gap functions through closest-point or mortar projections on the current surface [1, 2]. These formulations provide a direct route to contact sensitivities, but repeated projection at many slave samples can dominate contact-query cost. Our method keeps projection as an internal grid-population kernel and moves repeated queries to a current-space SDF field.

Distance fields and level sets provide a geometric representation in which proximity and normals are queried from a scalar field [3, 4, 5]. They have also been used in collision and contact pipelines in graphics and simulation [6]. In contrast to static distance fields, the field here is rebuilt from the current finite-element boundary and stores closest-feature payloads for master-side sensitivities.

Dynamic collision systems often use spatial subdivision, hashing, or bounding hierarchies to accelerate broad-phase search [7]. We use spatial hashing only inside field construction; the paper-facing contact query is interpolation of the narrow-band field. Learned implicit fields such as DeepSDF [8] demonstrate the value of continuous SDF representations, but our setting deliberately avoids snapshots, reduced bases, and neural training so that the distance field remains induced by the current FEM state.

3 FEM-Induced Dynamic Narrow-Band SDF Field

Let the current master boundary be a triangulated finite-element surface

$$\Gamma_h(\mathbf{q}) = \bigcup_e \mathbf{s}_e(\xi, \eta, \mathbf{q}), \qquad \mathbf{s}_e(\xi, \eta, \mathbf{q}) = \sum_{a \in I_e} N_a^\Gamma(\xi, \eta)\mathbf{x}_a. \tag{1}$$

At each field update, a Cartesian narrow band $\mathcal{G}_\rho(\Gamma_h)$ is constructed around the current surface. For each grid node $\mathbf{y}_\ell$, the field stores

$$(\Phi_\ell, \mathbf{d}_\ell, f_\ell^*, \boldsymbol{\xi}_\ell^*, \mathbf{n}_\ell, m_\ell), \tag{2}$$

where $\Phi_\ell = \phi_h(\mathbf{y}_\ell, \mathbf{q})$, $\mathbf{d}_\ell$ is an optional stored gradient or normal estimate, f_ℓ^* is the closest surface triangle, $\boldsymbol{\xi}_\ell^*$ are barycentric coordinates on that triangle, $\mathbf{n}_\ell$ is the current closest-feature normal, and m_ℓ is the narrow-band validity mask. The contact derivative is not defined by interpolating $\mathbf{d}_\ell$; it is defined as the derivative of the scalar interpolation of Φ_ℓ.

Closest-point projection is used here as a field construction kernel. It is not the main contact query. The implementation rejects out-of-band or invalid interpolation cells rather than silently falling back to projection.

4 Field-Interpolated Contact Formulation

For a slave sample point

$$\mathbf{x}_i^A = \sum_{b \in I_A} N_b^A \mathbf{x}_b^A, \tag{3}$$

the contact gap is the trilinearly interpolated SDF:

$$g_i = \hat{\phi}_B(\mathbf{x}_i^A) = \sum_{\ell \in \mathcal{C}(\mathbf{x}_i^A)} w_\ell(\mathbf{x}_i^A)\Phi_\ell. \tag{4}$$

The spatial derivative used by the slave Jacobian is the analytic derivative of this scalar interpolation:

$$\nabla_x \hat{\phi}_B(\mathbf{x}_i^A) = \sum_{\ell \in \mathcal{C}(\mathbf{x}_i^A)} \Phi_\ell \, \nabla_x w_\ell(\mathbf{x}_i^A). \tag{5}$$

The contact normal is the normalized scalar-field derivative,

$$\mathbf{n}_i = \frac{\nabla_x \hat{\phi}_B(\mathbf{x}_i^A)}{\|\nabla_x \hat{\phi}_B(\mathbf{x}_i^A)\|}. \tag{6}$$

An interpolated closest-feature normal $\tilde{\mathbf{n}}_i = \mathrm{normalize}(\sum_\ell w_\ell \mathbf{n}_\ell)$ is retained only as a geometric normal estimate for diagnostics or explicit refinement; it is not the derivative of the scalar gap. The sign convention is positive for separation and negative for penetration.

The slave Jacobian is

$$\frac{\partial g_i}{\partial \mathbf{x}_b^A} = N_b^A \nabla_x \hat{\phi}_B(\mathbf{x}_i^A)^T. \tag{7}$$

The master Jacobian is assembled from the interpolated closest-feature payload:

$$\frac{\partial g_i}{\partial \mathbf{x}_a^B} = -\sum_{\ell \in \mathcal{C}(\mathbf{x}_i^A)} w_\ell(\mathbf{x}_i^A) N_a^\Gamma(\boldsymbol{\xi}_\ell^*) \mathbf{n}_\ell^T. \tag{8}$$

For frictionless normal penalty contact,

$$E_c = \frac{1}{2} k \langle -g_i \rangle_+^2, \qquad \lambda_i = k \langle -g_i \rangle_+, \qquad \mathbf{f}_{c,i} = \mathbf{J}_i^T \lambda_i. \tag{9}$$

A Gauss-Newton contact stiffness approximation is assembled as $k\mathbf{J}_i^T \mathbf{J}_i$.

5 Algorithm and Cost Model

One update/query cycle extracts current boundary triangles, builds $\mathcal{G}_\rho(\Gamma_h)$, populates grid-node distance and payload values with the projection kernel, evaluates contact samples by interpolation, differentiates the scalar interpolation for slave sensitivities, and assembles field-contact Jacobians and penalty forces.

The field is beneficial only when the update cost is amortized over enough queries. With optional refinement count Q_r, the measured gate is

$$T_{\text{update}} + QT_{\text{field}} + Q_r T_{\text{refine}} < QT_{\text{projection}}. \tag{10}$$

Without refinement,

$$Q^* = \frac{T_{\text{update}}}{T_{\text{projection}} - T_{\text{field}}}. \tag{11}$$

The paper therefore claims acceleration only for query counts exceeding the measured crossover Q^*.

Table 1: Dynamic narrow-band SDF field accuracy on the nonplanar current surface.

Spacing	Grid nodes	Valid nodes	Max ϕ error	Max normal error	Max Eikonal residual
0.100	1792	876	1.835893×10^{-3}	5.751473×10^{-2}	6.784055×10^{-3}
0.075	3969	2174	1.132281×10^{-3}	4.769023×10^{-2}	4.546733×10^{-3}
0.050	12493	7826	5.114950×10^{-4}	3.600289×10^{-2}	4.743522×10^{-3}

6 Implementation

The field implementation is in `src/sfc/sdf/narrow_band_grid.py` and `src/sfc/sdf/dynamic_narrow_band_sdf.py`. The query API includes `query_phi` and `query_spatial_derivative_phi`; `query_gradient` is the derivative of the scalar ϕ interpolation, while `query_geometric_normal` exposes the optional interpolated closest-feature normal. The field-contact implementation is in `src/sfc/contact/field_contact.py`. It contains a preserved `node_to_surface_field_penalty_response` path for node/sample contact, a reference `surface_to_surface_field_penalty_response` path for triangle quadrature, and a vectorized surface-to-surface response used for the paper-facing CalculiX native-contact comparisons. The older projection path in `src/sfc/sdf/dynamic_surface_sdf.py` is retained as `surface_projection_distance_kernel` for field construction, validation references, and explicitly requested refinement.

The core finite-element mechanics remain internally assembled and independent of Abaqus-generated matrices, force files, `.odb` files, and spreadsheet outputs. No friction, self-contact, nonlinear FEM main method, GPU path, barrier contact, POD, neural SDF, or Abaqus dependency is introduced.

7 Experiments

The primary SDF-field outputs are generated by:

```
python validation/run_true_dynamic_sdf_field_validation.py
--out-dir results/true_sdf_final
```

The experiment uses a deterministic nonplanar triangulated current surface with 512 triangles and three grid spacings. The query path is interpolation-only; projection appears only in field construction and reference/error measurement. All reported gradient and Eikonal quantities use $\sum_\ell \Phi_\ell \nabla_x w_\ell$, the derivative of the scalar gap interpolation.

7.1 Field Accuracy and Eikonal Residual

This experiment evaluates whether the rebuilt current-space field remains a usable local SDF as grid spacing changes. The reference distance and normal are computed by projecting each sample to the current triangulated surface, while the method evaluates the same samples by field interpolation. Table 1 shows that the scalar gap error decreases as the spacing is refined; the normal error remains bounded because it is computed from the piecewise trilinear scalar derivative on a triangulated nonplanar surface.

7.2 Field Contact Jacobian and Force

This experiment checks whether the field-contact Jacobian is actually the derivative of the field gap used in the penalty energy. The slave block is compared against finite differences of $\hat{\phi}(\mathbf{x})$, and

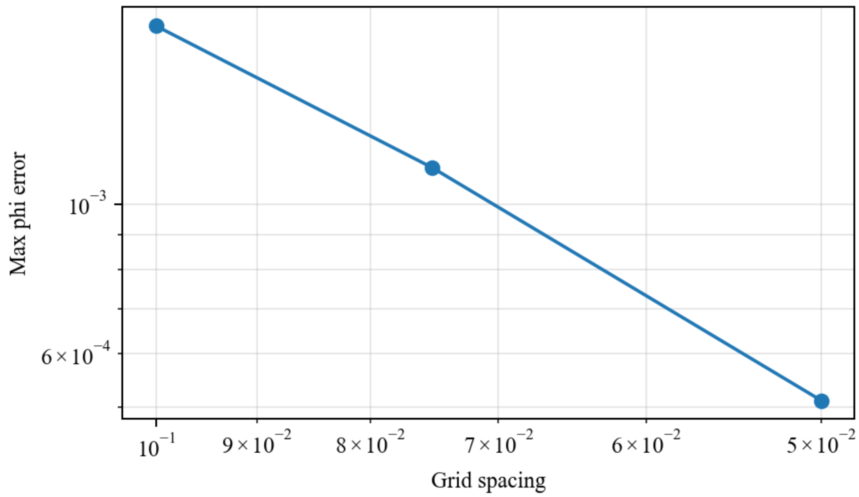

Figure 1: Field ϕ error decreases with grid spacing for the tested nonplanar surface.

Table 2: Field-contact Jacobian finite-difference check.

Case	Gap	Slave FD error	Master FD error	Status
single slave / large triangle	-1.30×10^{-1}	2.875566×10^{-11}	1.462164×10^{-11}	passed

the master block is compared against finite differences after rebuilding the field under master nodal perturbations. The small errors in Table 2 verify both the scalar-interpolation derivative used on the slave side and the closest-feature payload sensitivity used on the master side.

7.3 Material-Space SDF Baseline

This experiment isolates why a current-space field is needed for deformable FEM contact. The baseline keeps a frozen material-space plane SDF and evaluates it after tilt or shear deformation, while the proposed field is rebuilt from the current surface. The material-space field accumulates gap and normal errors because it no longer represents the deformed boundary; the rebuilt dynamic field avoids this mismatch.

7.4 Amortized Timing and Crossover

The dynamic field is not claimed to be faster for small query counts. Its update cost is paid once per rebuild, while the interpolation query cost is paid per contact sample. Table 4 reports the measured crossover Q^*: finer spacing improves field resolution but increases update cost, so Q^* rises from hundreds to several thousand queries.

The field acceleration claim is supported only for $Q > Q^*$ under the measured inequality. Optional refinement is implemented and timed, but the main acceleration gate uses $Q_r = 0$.

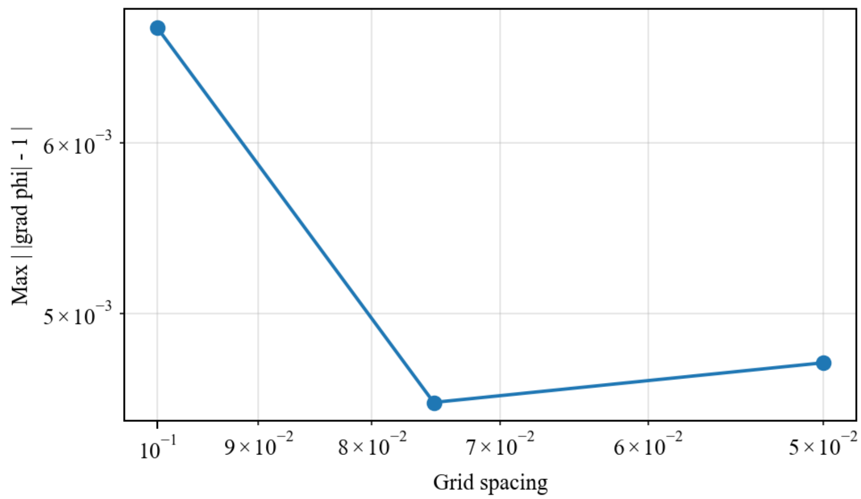

Figure 2: Maximum Eikonal residual over the dynamic narrow band.

Table 3: Frozen material-space SDF compared with the rebuilt dynamic field.

Case	Dynamic ϕ err.	Dynamic normal err.	Material ϕ err.	Material normal err.
tilt_x	8.326673×10^{-17}	1.523921×10^{-15}	9.783015×10^{-2}	1.193580×10^{-1}
shear_xy	5.551115×10^{-17}	3.351421×10^{-15}	4.265482×10^{-2}	9.962740×10^{-2}

7.5 Native Contact Solver Comparison

We further compare the independent SFC true-field contact solve against CalculiX native surface-to-surface contact on three benchmark-style contact geometries. CalculiX receives its own `*CONTACT PAIR` model and SFC independently solves the corresponding dynamic-SDF contact model; the comparison is not a replay and does not use CalculiX inside the core SDF path. Table 5 reports the largest displacement, strain, stress, and von Mises relative errors over the load steps. All rows pass the configured 5% native-solver agreement gate.

7.6 Solver-Level Timing and Backend Ablation

The field-query crossover is also reflected at the step level in contact-dominated TET4/HEX8 static and dynamic tests. These tests compare the true-field path against the spatial-hash projection timing reference while keeping the gap and force agreement at numerical roundoff. The field path is faster in all 16 non-quick rows, with measured projection/field step speedups from 7.51× to 24.17× and a minimum measured query crossover of 24.61 samples for the tested workload.

Finally, a backend ablation separates the preserved node-to-surface path from the surface-to-surface implementations. The non-vectorized surface-to-surface path is a correctness reference; the vectorized path preserves the same endpoint metrics while reducing the quadrature/contact query cost. Forced spatial-hash grouped field construction is retained for larger candidate sets, although it is slower on this small representative ablation because grouping overhead dominates.

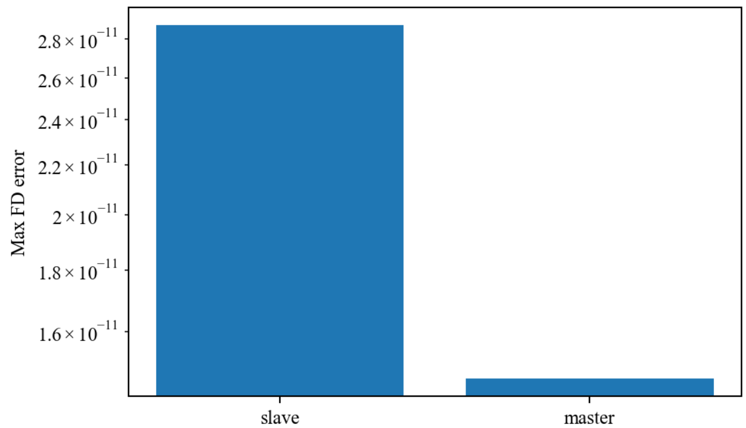

Figure 3: Slave and master field-contact Jacobian finite-difference errors.

Table 4: Update/query timing and measured crossover count. Times are seconds.

Spacing	T_{update}	T_{field}	$T_{\text{projection}}$	T_{refine}	Q^*
0.100	14.330253	5.626483×10^{-5}	1.749774×10^{-2}	5.731850×10^{-4}	822
0.075	29.062764	5.340050×10^{-5}	1.747250×10^{-2}	5.369825×10^{-4}	1669
0.050	78.940208	5.365200×10^{-5}	1.742862×10^{-2}	4.956188×10^{-4}	4544

7.7 External Visual FEM Validation

The preceding experiments verify the dynamic SDF field itself: field accuracy, Eikonal residuals, field-contact Jacobians, and the amortized query crossover. To make the resulting FEM/contact fields visually auditable, we retain a generated three-dimensional boundary-surface comparison to exercise the same visualization artifact format without requiring any external solver. This diagnostic comparison is not used to prove the SDF acceleration claim; it checks whether displacement, stress, strain, gap, and contact-pressure fields are physically interpretable in a controlled configuration. CalculiX-style external data can be substituted in the same artifact format, but no external solver is required by the core SDF field construction or contact query.

We also use open external reference lines rather than closed benchmark material without public numeric targets. SfePy's public two-body contact example solves a penalty-contact state with an independent finite-element code path [9]; SFC then rebuilds a current-space `DynamicNarrowBandSDF` field from the final master surface and replays the slave samples using interpolation-only `query_phi` calls. The quick external replay in Table 6 gives matching contact activation and sub-1.2×10^{-3} mean-gap differences for the tested approach values. The FuzzyContact Mendeley dataset [10] is retained as an optional VTU field-cloud source because it publishes displacement, strain, and stress components for several contact scenarios; it is not downloaded by default and is not a core dependency.

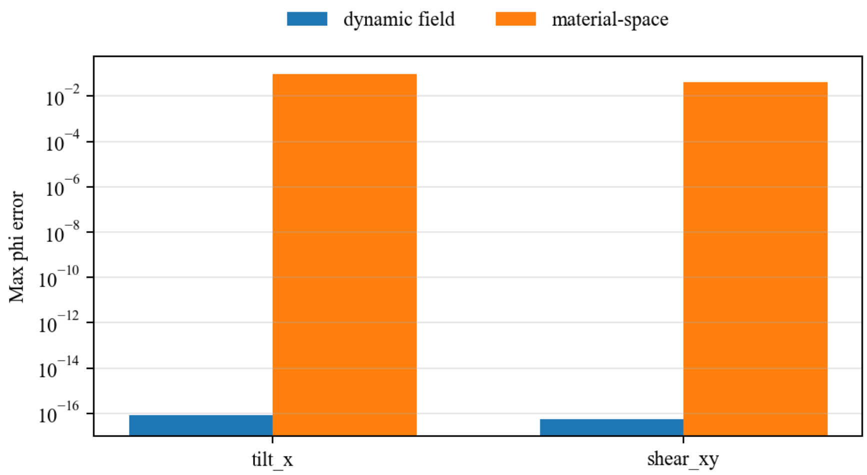

Figure 4: Material-space SDF error under current-surface deformation compared with the rebuilt dynamic field.

Table 5: Independent CalculiX native contact versus SFC dynamic-SDF contact. Times are wall-clock seconds for the tested Python/CalculiX runs.

Problem	Disp. err.	Strain err.	Stress err.	VM err.	SFC	CCX	CCX/SFC
sphere indentation	1.380272×10^{-2}	1.805590×10^{-2}	1.630772×10^{-2}	1.599641×10^{-2}	1.870	2.872	1.536
V-indenter	2.301961×10^{-2}	3.961379×10^{-2}	3.697091×10^{-2}	3.304457×10^{-2}	0.628	3.778	6.015
cylinder compression	6.558312×10^{-4}	2.191263×10^{-3}	1.725421×10^{-3}	1.294581×10^{-3}	12.678	9.343	0.737

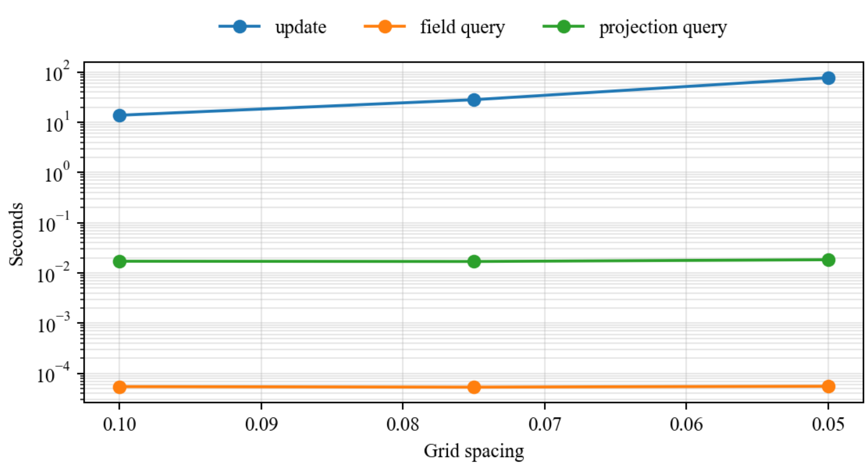

Figure 5: Field update, field query, and projection query timing components.

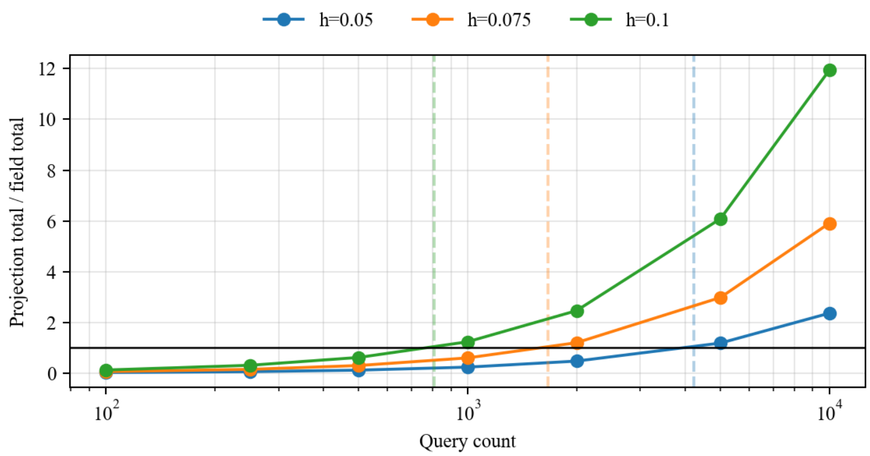

Figure 6: Amortized speedup as projection-query total time divided by SDF-field total time. Dashed vertical markers indicate measured crossover counts.

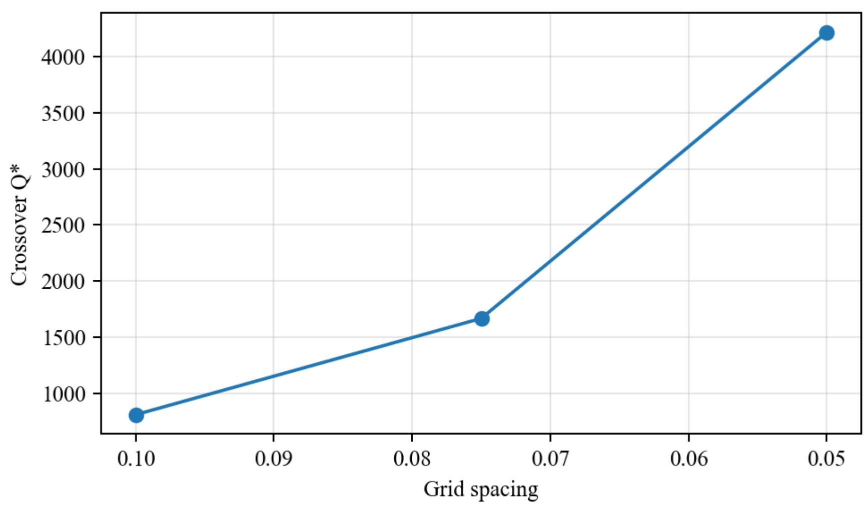

Figure 7: Measured crossover query count Q^* for each grid spacing.

Table 6: Open SfePy two-body contact replay with SFC true-field interpolation queries. SfePy solves the external contact state; SFC only rebuilds the dynamic SDF field on the final geometry and queries it by interpolation.

Approach	SfePy active	SFC active	Mean-gap diff.
0.00	0	0	1.000000×10^{-4}
0.06	2	1	7.119386×10^{-4}
0.10	2	1	1.187357×10^{-3}

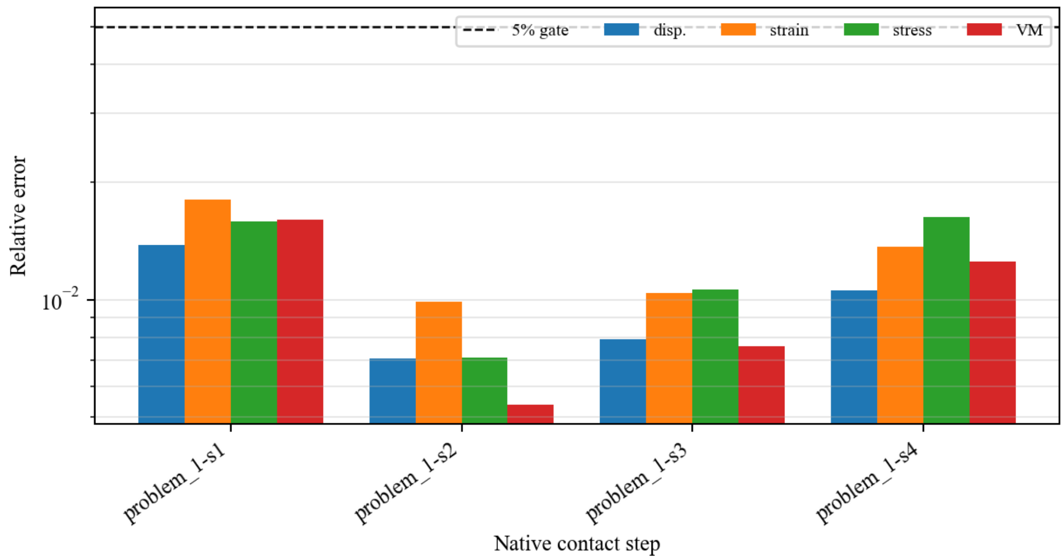

Figure 8: Native-contact agreement errors for the sphere-indentation case. The same figure set is generated for the V-indenter and cylinder-compression cases; Table 5 summarizes all three.

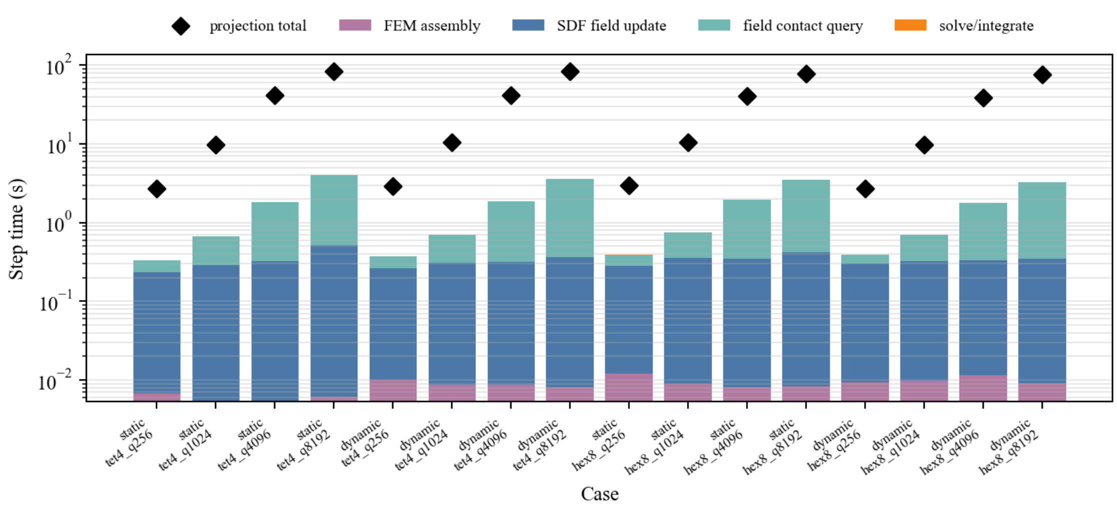

Figure 9: Solver-level timing breakdown for TET4/HEX8 static and dynamic contact-dominated workloads. Diamonds mark the projection-reference total step time.

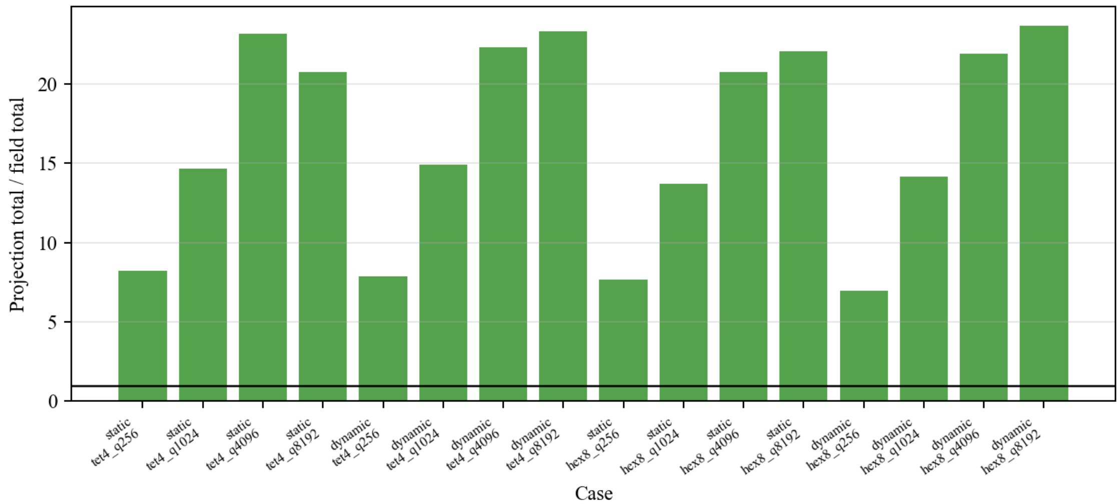

Figure 10: Projection-reference total step time divided by true-field total step time for the solver-level timing cases.

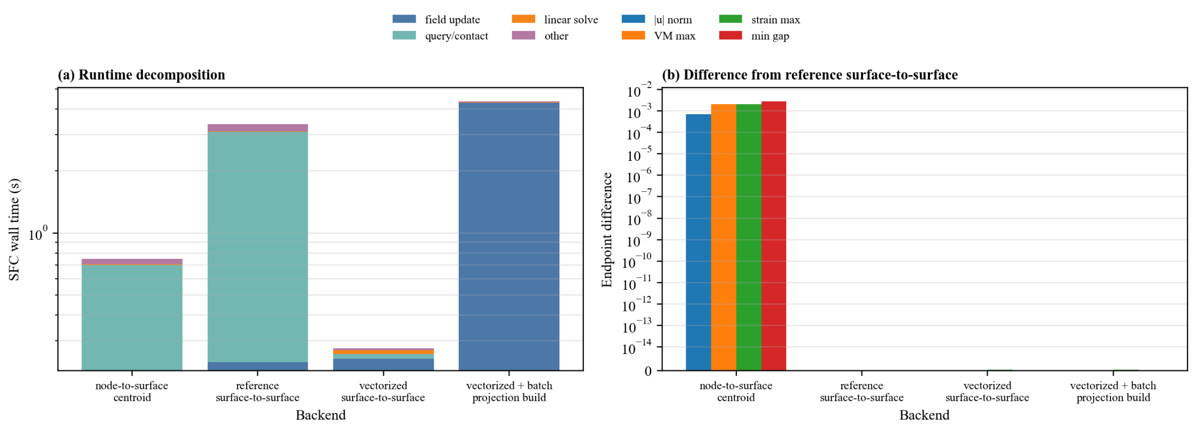

Figure 11: Representative backend ablation for node-to-surface sampling, reference surface-to-surface quadrature, vectorized surface-to-surface quadrature, and vectorized field contact with forced batch projection during field construction. The legend is placed outside the axes so the timing bars and endpoint-difference bars remain unobstructed.

Table 7: External visual validation metrics for the generated reference case. The comparison is diagnostic and supports physical plausibility, not SDF acceleration.

Metric	Value
Displacement L2 relative error	1.404040×10^{-2}
von Mises L2 relative error	2.699310×10^{-2}
Strain-norm L2 relative error	2.104498×10^{-2}
Gap L_∞ absolute error	4.000000×10^{-4}
Pressure L2 relative error	9.654852×10^{-3}
Contact-force relative error	8.775744×10^{-3}

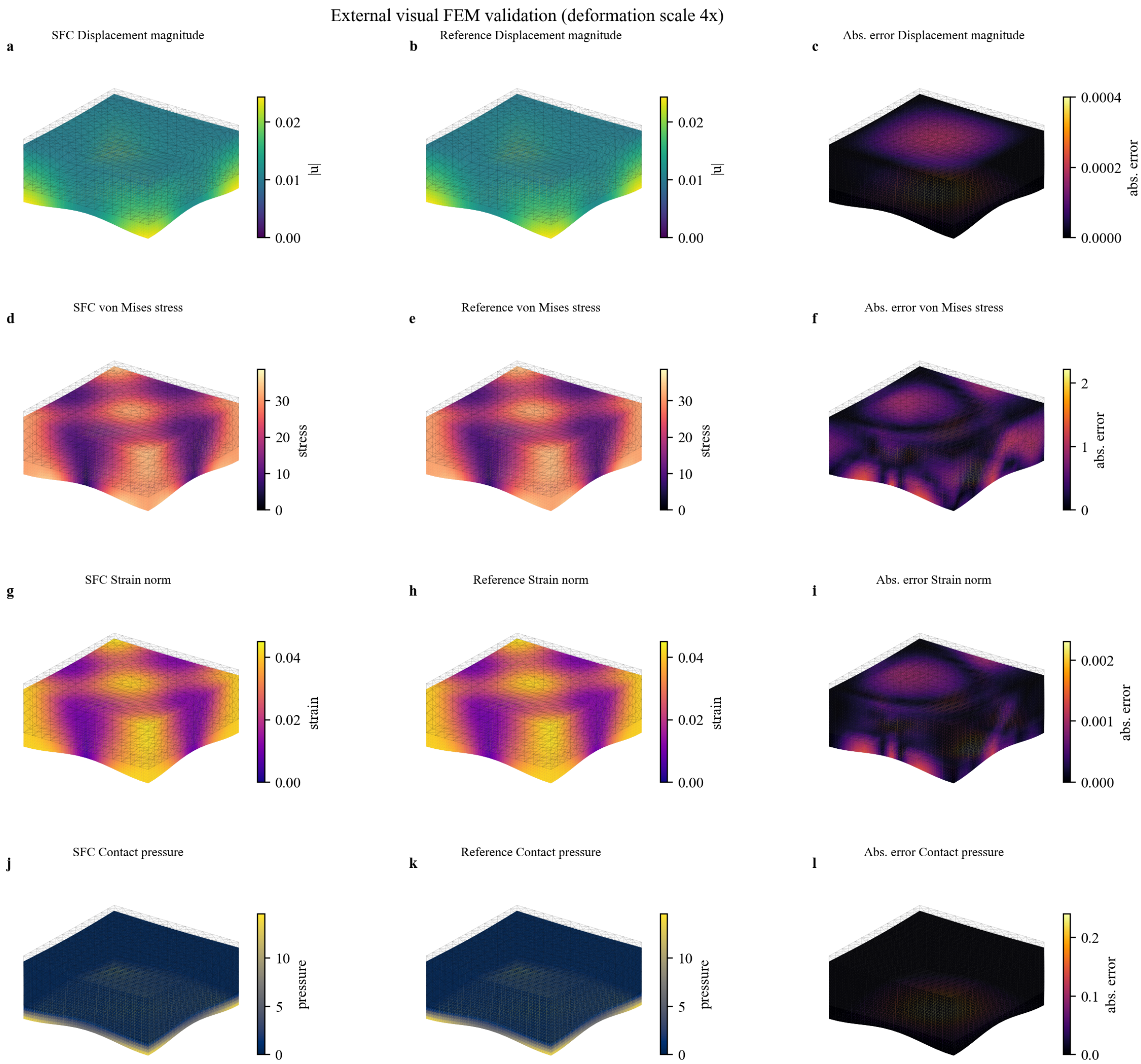

Figure 12: External visual FEM validation rendered on the deformed three-dimensional boundary surface. Each row compares SFC, generated reference, and absolute error with matched SFC/reference color scales and a deformation scale of 4×.

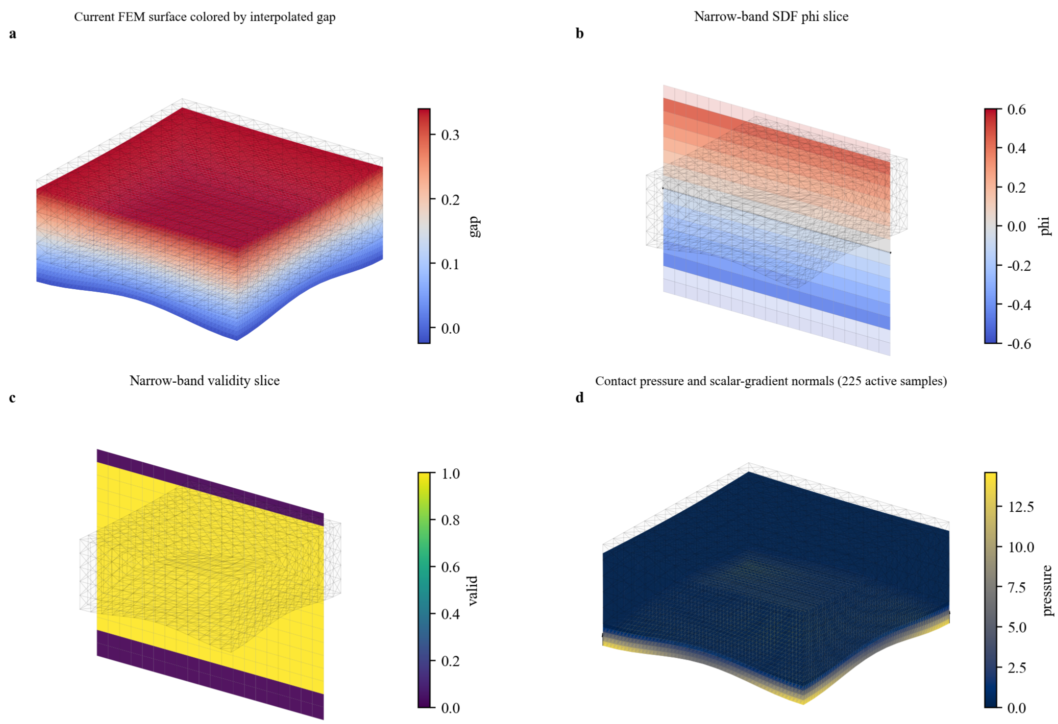

Figure 13: Dynamic SDF field visualization rendered as three-dimensional surfaces: current FEM boundary colored by interpolated gap, narrow-band ϕ slice, narrow-band validity slice, and contact-pressure surface with scalar-gradient normals.

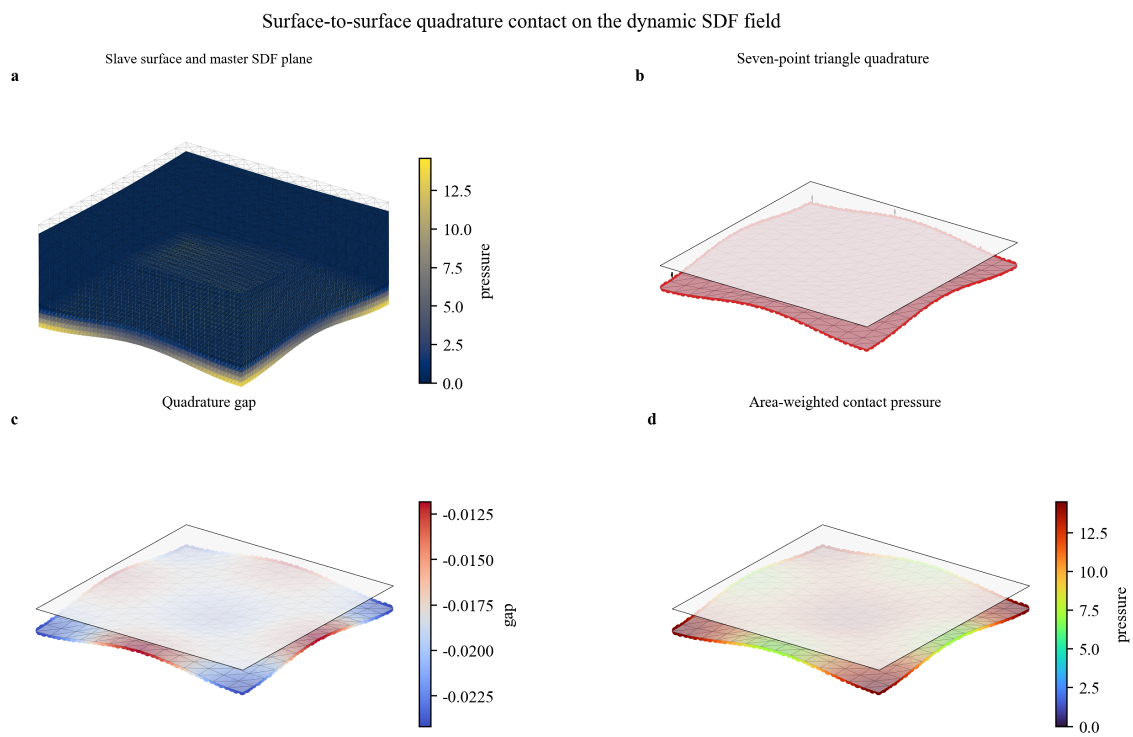

Figure 14: Surface-to-surface quadrature contact visualization. The panels show the deformed slave surface, the master SDF plane, seven-point triangle quadrature samples, interpolated gaps, area-weighted contact pressure, and scalar-field normals.

8 Scope of Claims

The reported experiments support true dynamic narrow-band SDF field existence, interpolation-only query, field accuracy under the reported thresholds, slave/master field-contact Jacobian finite-difference consistency, and SDF acceleration after the measured Q^* crossover. The solver-level tests support the scoped statement that the optimized true-field backend can reduce contact-dominated TET4/HEX8 step time after the measured crossover, while the backend ablation documents both node-to-surface and surface-to-surface integration paths.

The external visual validation supports physical auditability of the tested displacement, stress, strain, gap, and pressure fields. It is not evidence for the SDF acceleration claim and is not a claim of external-solver equivalence.

The SfePy open-reference replay supports a scoped external contact-state check for the tested two-body penalty case. It does not convert SfePy into a dependency and does not establish full nonlinear contact equilibrium equivalence. The FuzzyContact line currently supports open VTU dataset availability and field-cloud parsing, with quantitative solver comparison deferred until a specific downloaded case is mapped to SFC replay samples.

The current results do not support claims of robust global SDF behavior for arbitrary non-manifold geometry, friction, self-contact, nonlinear FEM as the main method, GPU acceleration, barrier contact, superiority over production BVH, or Abaqus-dependent core solver behavior.

9 Supplementary Context

Additional CalculiX, C3D8, and external-solver replay results are not the main evidence for this method. They may be retained as supplementary context for external deformed-state checks, but the main paper claims are field construction, field interpolation, field-contact Jacobians, penalty force assembly, and amortized SDF update/query cost.

10 Limitations

The field is valid only in the constructed narrow band. Sign handling assumes consistently oriented current surfaces in the tested cases. The current paper-facing implementation targets internally assembled linear finite-element mechanics and frictionless normal penalty contact. Timing results are Python prototype measurements and should not be interpreted as production BVH or hardware acceleration claims.

11 Conclusion

This paper presented a training-free dynamic narrow-band SDF field for finite-element contact. The key distinction from projection-query contact is that closest-point projection is internalized as a field construction kernel, while repeated contact evaluation is performed by interpolation of Φ_ℓ and closest-feature payloads. The slave sensitivity is the analytic derivative of the scalar SDF interpolation, and the master sensitivity is assembled from closest-feature payloads. The experiments show field accuracy, Eikonal consistency, finite-difference Jacobian agreement, material-space SDF failure under deformation, and measured crossover counts for amortized acceleration.

Data Availability

The scripts and tabulated numerical data used for the reported validation studies are available from the corresponding author upon reasonable request. A public archival repository can be linked at submission if required by the journal or funding body.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used OpenAI Codex to support manuscript organization, consistency checks, and language revision. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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